Parameter	Name	Value	Units	Source
Ra_c / Ra_C	Critical Rayleigh Number	1000	—	Mulyukova & Bercovici 2020, cites: Chandrasekhar, S. (1961). *Hydrodynamic and hydromagnetic stability*. Oxford, U.K.: Oxford University Press.
Rem_c / Rem_C	Critical Magnetic Reynolds Number	40	—	–
age_SS	Age of the Solar System	4.5	Gyr	–
m_neptune	Mass of Neptune	102.409e24	kg	https://nssdc.gsfc.nasa.gov/planetary/factsheet/neptunefact.html
a / a_triton	Semi-Major Axis of Triton	354.76e3	km	https://nssdc.gsfc.nasa.gov/planetary/factsheet/neptuniansatfact.html
Omega / omega_triton	Angular Rotation Rate of Triton	1.237432e-5	rad / s	2π/T, T from https://nssdc.gsfc.nasa.gov/planetary/factsheet/neptuniansatfact.html
c_p / cp_mean	Mean Specific Heat Capacity of Triton	1000	J / kg / K	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
P_cmb / P_CMB	Pressure at Triton's Core/Mantle Boundary	1.9	GPa	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
T_S / T_OMB	Temperature at Triton's "Surface" (Ocean/Mantle Boundary)	250	K	–
r_core / R_C	Radius of Triton's Core	425	km	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
r_mantle / R_M	Radius of Triton's Mantle	1028	km	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
r_ocean / R_O	Radius of Triton's Ocean	1111	km	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
r_ice / R_I	Radius of Triton's Ice Shell	1353	km	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
rho_core / rho_C	Density of Triton's Core	6	g / cm^3	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
rho_mantle / rho_M	Density of Triton's Mantle	3.3	g / cm^3	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
rho_ocean / rho_O	Density of Triton's Ocean	1	g / cm^3	Pure water, not specified in Nimmo et al. 2025.
rho_ice / rho_I	Density of Triton's Ice Shell	0.95	g / cm^3	Nimmo, F., Bierson, C., McKinnon, W. B. (2025) Pluto and Triton: interior structures, lithospheres, and potential for oceans, in *Triton and Pluto: the long lost twins of active worlds*, edited by Luspay-Kuti, A. and Mandt, K., pp. 2.1-2.31, IOP Publishing, Bristol.
initial_iso['U_238'] / U_238_props	Properties of ^{238}U (Half-Life, Heat Production per Unit Mass, Natural Abundance)	[4.46e9, 95.13, 99.27]	[yr, µW/kg, %]	Jaupart et al. 2015 cites: S.T. Dye Geoneutrinos and the radioactive power of the Earth Reviews of Geophysics, 50 (2012), p. 3007
initial_iso['U_235'] / U_235_props	Properties of ^{235}U	[7.04e8, 568.47, 0.72]	[yr, µW/kg, %]	Jaupart et al. 2015 cites: S.T. Dye Geoneutrinos and the radioactive power of the Earth Reviews of Geophysics, 50 (2012), p. 3007
initial_iso['Th_232'] / U_238_props	Properties of ^{232}Th	[1.40e10, 26.3, 100]	[yr, µW/kg, %]	Jaupart et al. 2015 cites: S.T. Dye Geoneutrinos and the radioactive power of the Earth Reviews of Geophysics, 50 (2012), p. 3007
initial_iso[K_40'] / K_40_props	Properties of ^{40}K	[1.26e9, 28.47, 0.0117]	[yr, µW/kg, %]	Jaupart et al. 2015 cites: S.T. Dye Geoneutrinos and the radioactive power of the Earth Reviews of Geophysics, 50 (2012), p. 3007
T_P_init / T_M_i	Initial Temperature of Adiabatic Gradient Projected to the Top of the Mantle	1800 (at t = 0)	K	–
T_C_init / T_C_i	Initial Temperature at Triton's Core/Mantle Boundary	2200 (at t = 0)	K	–
k / k_M	Thermal Conductivity	4.2	W / m / K	Castillo-Rogez & Lunine 2010, cites: Grindrod, P. M., A. D. Fortes, F. Nimmo, D. L. Feltham, J. P. Brodholt, and L. Vočadlo (2008), The long-term stability of a possible aqueous ammonium sulfate ocean inside Titan, *Icarus*, **197**, 137–151. * Find specifically where this value is from, could also possibly be from: Clauser, C., and E. Huenges (1995), Thermal conductivity of rocks and minerals, in *Rock Physics and Phase Relations: A Handbook of Physical Constants, AGU Ref. Shelf*, vol. **3**, edited by T. J. Ahrens, pp. 105–126, AGU, Washington, D. C.
c_p_mantle / cp_M	Heat Capacity of Triton's Mantle	900	J / kg / K	Castillo-Rogez & Lunine 2010, cites: Grindrod, P. M., A. D. Fortes, F. Nimmo, D. L. Feltham, J. P. Brodholt, and L. Vočadlo (2008), The long-term stability of a possible aqueous ammonium sulfate ocean inside Titan, *Icarus*, **197**, 137–151. * Find specifically where this value is from.
eta_0 / eta0_M	Reference Viscosity (for Dynamic Viscosity)	1e20	Pa s	–
E / E_M	Activation Energy	3e5	J / mol	–
T_0 / T0_M	Reference Temperature (for Dynamic Viscosity)	1600	K	–
kappa / kappa_M	Thermal Diffusivity	1e-6	m^2 / s	–
alpha / alpha_M	Thermal Expansivity	3e-5	1 / K	Castillo-Rogez & Lunine 2010, cites: Fei, Y. (1995), Thermal expansion, in *Rock Physics and Phase Relations: A Handbook of Physical Constants, AGU Ref. Shelf*, vol. **3**, edited by T. J. Ahrens, pp. 29–44, AGU, Washington, D. C. * Find specifically where this value is from.
k_core / k_C	Thermal Conductivity of Triton's Core	5	W / m / K	–
eta_m / etaM_C	Magnetic Diffusivity of Triton's Core	2	m^2 / s	–
gamma / gamma_C	Constant (for Dipole Moment)	0.15	—	Olson & Christensen 2006
mu_0 / mu0_C	Magnetic Permeability	4e-7 * pi	H / m	Olson & Christensen 2006
wt_S / wtS_C	Weight Percent of Sulfur in Triton's Core	12.506	%	–
ppm_U / ppmU_M	Uranium Concentration (CI Chondrites)	0.008	ppm	Jaupart et al. 2015 ("Temperatures, Heat, and Energy in the Mantle of the Earth") cites: H. Palme, H.S.C. O'Neill Cosmochemical estimates of mantle composition R.W. Carlson (Ed.), Mantle and Core, Treatise on Geochemistry, vol. 2, Elsevier, Amsterdam (2003)
ppm_Th / ppmTh_M	Thorium Concentration (CI Chondrites)	0.030	ppm	Jaupart et al. 2015 cites: H. Palme, H.S.C. O'Neill Cosmochemical estimates of mantle composition R.W. Carlson (Ed.), Mantle and Core, Treatise on Geochemistry, vol. 2, Elsevier, Amsterdam (2003)
ppm_K / ppmK_M	Potassium Concentration (CI Chondrites)	544	ppm	Jaupart et al. 2015 cites: H. Palme, H.S.C. O'Neill Cosmochemical estimates of mantle composition R.W. Carlson (Ed.), Mantle and Core, Treatise on Geochemistry, vol. 2, Elsevier, Amsterdam (2003)
ppm_U_core / ppmU_C	Uranium Concentration of Triton's Core	0	ppm	–
ppm_Th_core / ppmTh_C	Thorium Concentration of Triton's Core	0	ppm	–
ppm_K_core / ppmK_C	Potassium Concentration of Triton's Core	10	ppm	–
leaching_coeff	^{40}K Ocean Leaching Coefficient (0: ^{40}K has fully leached into ocean, 1: ^{40}K has not leached into ocean at all)	1	—	–